The Addition of HCl to Carvone

Name: Key

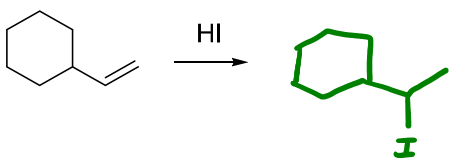
1) What is the name of the **reverse reaction** of an HX (e.g. HBr) addition to an alkene? (1 pt)

Elimination reaction (E1)

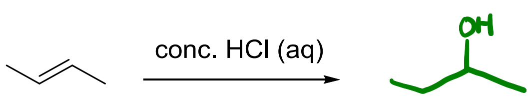
2) Is the addition of H-Br to an alkene entropically favored or disfavored? Explain why in 1-2 sentences. (2 pts)

Entropically disfavored; the reaction starts with 2 molecules (alkene + HBr) and ends with one molecule (addition product), therefore decreasing the disorder of the reaction (- delta S).

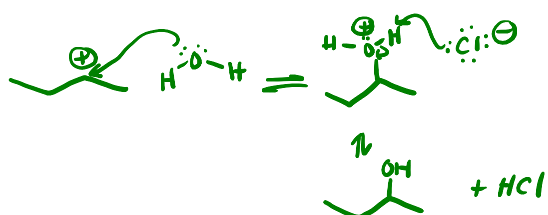
3) What is the **major product** formed from the addition of HI to vinylcyclohexane (shown below)? (2 pts)



4) A chemist attempts to chlorinate *E*-2-butene using concentrated, aqueous HCl solution fails to isolate any 2-chlorobutane. What undesired product would be formed instead? (2 pts)



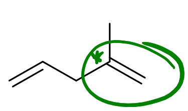
The water from the aqueous HCl is nucleophilic and can attack the carbocation once it is formed.



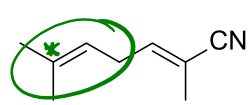
5) The organic compounds below each contain two alkenes. In each example:

- Circle the individual alkene that would **react fastest** with an equivalent of HCl. (2 pts)
- Within the alkene that reacts the fastest, use an asterisk (*) to indicate which carbon chlorine would add to. (2 pts)

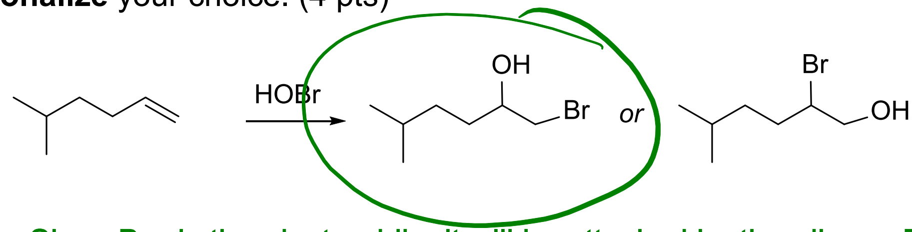
a)



b)



6) Alkenes react with hypobromous acid (HOBr) to generate halohydrins and proceed through a very similar mechanism as HX additions (Hint: Br is the electrophile in this reaction). Although two regioisomers are possible, only one is formed. Select the **major isomer** and **rationalize** your choice. (4 pts)



Since Br+ is the electrophile, it will be attacked by the alkene. The resulting carbocation can be either secondary (more stable) or primary. The secondary carbocation will form in much larger amounts and will be attacked by the OH- nucleophile.